



Syllabus Application



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CS 305

Programming Languages

Faculty Faculty of Engineering and Natural Sciences

Semester Spring 2025-2026

Course CS 305 - Programming Languages

Time/Place	<u>Time</u>	<u>Week Day</u>	<u>Place</u>	<u>Date</u>
	13:40-14:30	Mon	UC-G030	Feb 16-May 22, 2026
	08:40-10:30	Fri	UC-G030	Feb 16-May 22, 2026

Level of course Undergraduate

Course Credits SU Credit:3, ECTS:6, Engineering:6

Prerequisites CS 204

Corequisites

Course Type Lecture

Program Requirements >

Instructor(s) Information

Hüsnü Yenigün

Email: yenigun@sabanciuniv.edu

Office Hours (by appointment): Tuesday 18:30-20:30

Please [reserve a slot](#) before 23:59PM of the previous day, latest.

Course Information

Catalog Course Description

This course discusses the concepts underlying the design, definition, implementation, and use of modern programming languages for a variety of programming paradigms, such as imperative, functional, logic and concurrent programming paradigms. The syntax and semantics of programming languages are described using formal approaches, and basic concepts of language analysis and compilation are covered using modern tools.

Course Learning Outcomes:

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|----|---|
| 1. | The students are expected to have basic information for the implementation of programming languages. |
| 2. | The students are expected to have some experience to apply techniques of language implementation by using tools for language implementation. |
| 3. | The students are expected to have basic information for the common concepts in programming languages. |
| 4. | The students are expected to have introductory information on programming language paradigms such as functional programming, logic programming, and parallel programming. |

Course Objective

To give the students the basic principles of various programming language paradigms and implementation of these languages.

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Course Materials

Resources:

TEXTBOOKS*:

- [1] "Programming Languages: Concepts and Constructs" by Ravi Sethi
- [2] "Concepts of Programming Languages" by Robert W. Sebesta
- [3] "Comparative Programming Languages" by Leslie B. Wilson and Robert G. Clark
- [4] "Programming Languages: Principles and Paradigms" by Allen Tucker and Robert Noonan
- [5] "Essentials of Programming Languages" by Daniel Friedman, Mitchell Wand, and Christopher

T. Haynes

*: A lecture notes document prepared based on the references above will be provided.

Technology Requirements:

You will need to have SSH access to cs305.sabanciuniv.edu server for homeworks. SU-Net users of the students will be activated on this server after the course starts. All the software packages you need for homeworks are already installed on this server.

Assessment

The final grade for this course will be based on:

	Percentage (%)	Number of assessment methods
Final	45%	
Midterm	40%	1
Homework	15%	5

Midterm Exam Date: April 12, 2026 @ 14:30-16:30

Final Exam Date: TBA

Similar to a failing threshold (which is decided at the end of the semester) for the overall numeric grade of the course, there will also be a failing threshold for the all the grading items (midterm, final, homeworks). If your grade from a grading item falls below the failing grade of a grading item, this will be a reason for failing the course. Again, similar to the failing threshold for the overall numeric grade of the course, failing thresholds for the midterm, final, homeworks will be decided at the end of the semester.

Exam Schedule:

<u>Time</u>	<u>Event</u>	<u>Details</u>
Apr 12, 2026 14:30	Midterm: CS305 Midterm	

Policies

Course Policies:

Make-up Policy: If you miss exactly one of the midterm or final exam, and if you have a valid excuse (e.g. a medical condition, an official university event participation, etc.), then you can take the make-up exam. In this case, the grade of the make-up exam counted as the grade of your missing exam. The make-up exam can be an oral exam, a written exam, or both.

AI Policies

[Guidelines on the "Use of Generative AI" for students](#)

SU Academic Integrity Statement:

<https://www.sabanciuniv.edu/en/academic-integrity-statement>

Health Report Requirements

[Student Medical Reports Instruction Letter](#) 

Weekly Content

Week	Topic
Week 1	Introduction, Describing Syntax and Semantics of Programming Languages
Week 2	Flex and Scanner Implementation
Week 3	Context Free Grammars
Week 4	Bison and Parser Implementation
Week 5	Abstract Syntax Trees, Semantic Analysis
Week 6	Expressions, Types and Type Checking, Statements, Scoping Rules
Week 7	Subprograms – Referencing Environments, Parameter Passing
Week 8	Subprograms – Activation Records
Week 9	Functional Programming – Expressions, Procedures
Week 10	Functional Programming – Data types
Week 11	Functional Programming – Interpreters
Week 12	Logic Programming – Relations, Rules/Facts, Inferencing

Week	Topic
Week 13	Logic Programming – Unifications, Programming Techniques
Week 14	Parallel Programming